

Open-NL: A Transparent Computational Testbed for Wide Dynamic Range Compression

Mark Shaver^{1, a)}

¹ Wichita State University, Department of Communication Sciences and Disorders, 1845 Fairmount St, Wichita, KS 67260, USA

^{a)}Email: mark.shaver@wichita.edu

Abstract

Open-NL is an open-source computational testbed designed for the transparent modeling and evaluation of Wide Dynamic Range Compression (WDRC) prescriptive rules. Modern prescriptions like NAL-NL2 rely on extensive empirical regularizations to ensure clinical safety. These safeguards counteract the well-documented failures of pure intelligibility maximization. However, clinical software remains closed-source. This prevents researchers from isolating the local impact of specific heuristic safeguards.

Open-NL addresses this limitation. It couples a multi-start Nelder-Mead Speech Intelligibility Index (SII) optimizer with the Moore & Glasberg (2004) specific-loudness model. To map the algorithmic behavior, we conducted two distinct experiments. First, the testbed was benchmarked across a Monte Carlo optimization stability sweep ($N = 20$ independent restarts per profile under aggressive $\mathcal{U}(-10, +10)$ dB noise). This confirmed that the multi-start Nelder-Mead simplex yielded robust convergence envelopes, proving the algorithm is structurally anchored by its objective boundaries rather than its initialization. Second, we

conducted a One-At-a-Time (OAT) parameter sensitivity analysis on the objective function’s
 four governing physiological limits. Because the objective function scales the base intelligibility
 score by 100 (i.e., $SII \times 100$) and enforces a linear penalty weight ($\lambda_{loud} = 2000$) on loudness
 growth, the mathematical break-even point for breaching the loudness cap by any increment
 Δ sones requires a compensatory raw intelligibility increase of $\Delta SII \geq \frac{2000}{100} \Delta = 20\Delta$. For any
 $\Delta \geq 0.05$, this requirement exceeds the entire dynamic range of the $[0, 1]$ index, establishing
 a virtually impenetrable optimization wall. The OAT sensitivity analysis confirms that
 the derivative-free solver successfully locks onto these mathematically dictated physiolog-
 ical boundaries (e.g., loudness ceilings and 3.0:1 Compression Ratios) without numerical
 divergence, proving that the constraints themselves, rather than the initial seed, dictate the
 optimization frontier.

Standard monaural models systematically underestimate real-world binaural broadband
 summation. Therefore, this modeled loudness frontier is illustrative rather than clinically
 definitive. Historically, unregularized optimization on profound sloping profiles collapses
 due to a fundamental structural phenomenon: “broadband loudness crowding-out,” where
 residual normal low-frequency hearing natively consumes the allowable physiological loudness
 budget, algebraically starving the profoundly impaired high-frequencies of audibility. Open-
 NL reveals that this entrapment cannot be resolved simply by reallocating gain within a strict
 0 dB insertion floor. Instead, Open-NL proves that to achieve theoretically optimal speech
 intelligibility for profound precipitous hearing losses without violating physiological loudness
 limits, active low-frequency attenuation is mathematically required. When permitted to
 utilize a realistic -10 dB vent-leakage floor, the optimizer aggressively suppresses normal-
 hearing low frequencies to artificially free up physiological capacity, allowing it to prescribe

substantially *more* high-frequency gain in the critical 2-4 kHz speech regions than standard empirical formulas. Open-NL exposes these critical trade-offs within a fully inspectable framework, demonstrating how algorithmic calibration can overcome the limits of pure mathematical optimization against real-world physiological bounds. Ultimately, it provides the computational substrate for future calibration workflows, paving the way to replace static heuristic boundaries with individualized, distortion-aware objective functions (Margolis et al., 2025).

I. INTRODUCTION

Manufacturer-agnostic prescriptions remain central to evidence-based hearing aid practice. While earlier investigations suggested that generic targets might outperform proprietary first-fit algorithms on patient preference and specific metrics (Valente et al., 2018), contemporary evidence indicates that aided speech recognition in noise often shows no significant difference across formulas (e.g., Cox et al., 2014). However, while formula choice has relatively modest intelligibility consequences in background noise, it drives substantial variations in overall loudness, making modeled loudness (quantified in sones per the Moore & Glasberg 2004 impaired loudness model) the primary dependent variable in prescriptive evaluation.

While the derivations of major algorithms like NAL-NL2 and DSL m[i/o] are published in detail, their software implementations remain closed-source. Audiological science has long recognized that unconstrained intelligibility maximization fails clinically without extensive empirical regularization. For example, the evolution from NAL-NL1 to NAL-NL2 required critical empirical corrections. These included global gain reductions and reduced compression

ratios for severe losses (Keidser et al., 2007). These safeguards were introduced specifically to counteract the aggressive over-amplification provoked by pure mathematical optimization (Keidser, Dillon, Carter, & O’Brien, 2012).

Because clinical fitting software packages are compiled black boxes, researchers cannot isolate individual heuristic rules. It remains impossible to observe the local sensitivity of specific safeguards within the optimization cascade. Existing open-source tools serve distinct, separate functional niches. The openMHA platform (Herzke et al., 2017) operates as a real-time signal processing master hearing aid rather than a target generator. The Cambridge CAM2/CAMEQ2-HF formulae (Moore et al., 2010) provide rigidly defined equation-based targets rather than a modular optimization sandbox. Finally, the Auditory Modeling Toolbox (AMT; Majdak et al., 2022) offers loudness modeling without native prescriptive inversion. Consequently, investigators cannot isolate a specific prescriptive heuristic within an optimization loop without reverse-engineering an entire proprietary engine.

Open-NL fills this gap. It provides a modifiable R substrate designed for the modular ablation of prescriptive heuristics. By coupling a Nelder-Mead desensitized SII optimizer to an integrated C++ specific-loudness engine, researchers can systematically disable, isolate, or invert individual heuristics. For instance, investigators can evaluate the upward spread of masking when disabling the 30 dB conductive safety cap. This modular architecture aligns directly with evolving audiological frameworks, such as the multi-profile philosophy introduced in NAL-NL3 (Kitterick et al., 2026).

To benchmark this testbed without introducing confounding variables, the optimization layer is embedded within a reproduced evaluation paradigm. The seven reference audiometric

profiles, the Moore & Glasberg (2004) specific-loudness model, and the ANSI S3.5 SII metric utilized herein are a direct replication of the methodological framework established by Johnson and Dillon (2011). (Throughout this manuscript, “ANSI SII” refers to raw physical audibility, whereas “smoothed desensitized SII” or “complete desensitized SII” refers to audibility incorporating severe-loss desensitization and level distortion penalties). Because this physiological evaluation space is already established in the literature, the primary contribution of this manuscript is the transparent computational testbed itself. By exposing the behavior of numerical solvers within this standardized sandbox, this manuscript clarifies a fundamental distinction: while numerical solvers converge stably on any fixed objective space, theoretical WDRC target generation exhibits acute parameter sensitivity to uncalibrated heuristic boundaries, providing the computational infrastructure necessary to evaluate local parameter robustness.

Clinical Safety and Usage Disclaimer

It is imperative to state unambiguously that Open-NL is a computational research testbed and **must not be used for fitting hearing aids on human listeners in its current form**. Because the framework deliberately permits aggressive, over-prescriptive targets for boundary testing—such as overriding empirical high-frequency roll-offs in profound losses to maximize intelligibility—it carries a significant risk of severe over-amplification. As established by Ching, Dillon, Katsch, and Byrne (2001), aggressive high-level targets in steeply sloping or profound losses are precisely where over-amplification risks are greatest, as the effectiveness of high-frequency audibility severely degrades as hearing loss worsens (desensitization). The hypotheses and targets generated by Open-NL represent extreme

mathematical boundaries intended to trigger experimental loudness rejection in controlled research settings, not clinical solutions. Any future behavioral translation of this framework requires independent institutional review, with mandatory real-ear verification and strict, individualized loudness-tolerance limits implemented as absolute prerequisites.

Reproducibility and Parametric Configuration

To ensure reproducibility and isolate the core WDRC optimization engine from confounding heuristic artifacts, all tables and figures reported in this manuscript were generated using a single, unified parameter vector: `optimize = TRUE`, `enable_severe_booster = FALSE`, `coupling = "bte_13"`, and crucially, a `-10 dB vent_floor` that explicitly permits realistic low-frequency physical attenuation. Disabling the severe-loss booster ensures that the extreme high-frequency gains observed in profiles A4 and A5 are the emergent mathematical product of the dynamic loudness cap and SII optimizer, rather than a hardcoded empirical booster. Furthermore, specifying the Behind-The-Ear (`bte_13`) coupling imposes realistic physical high-frequency roll-offs (`-15 dB` at `8 kHz`) natively into the objective function, preventing the solver from hallucinating physically impossible targets.

II. ALGORITHM ARCHITECTURE

A. Prescriptive Rationale and Objective Function

The choice of objective function is the primary design decision in any prescriptive formula. It governs the fundamental trade-off between intelligibility and comfort. Historically, established rationales occupy distinct positions on this spectrum. NAL-NL2 maximizes speech

intelligibility while constraining overall broadband loudness to be less than or equal to that of a normal-hearing listener (Keidser, Dillon, Carter, & O’Brien, 2012). Conversely, DSL m[i/o] normalizes loudness across frequency to restore normal dynamic range perception (Scollie et al., 2005). Finally, CAMEQ/CAM2 aim to equalize loudness across frequency bands (Moore, Glasberg, & Stone, 2010).

Open-NL positions its prescriptive rationale as a *constrained intelligibility-maximizer*. Its primary mathematical objective is the soft-constrained maximization of desensitized SII. Rather than globally restricting this maximization to a static “normal-or-less” loudness boundary, Open-NL restricts total sones using a dynamic, U-shaped broadband ceiling mapped across audiometric severity (PTA) via `cap_knots` (Section S.I.12; **Figure 1**). For example, at a 65 dB SPL input, the cap drops from 7.0 sones (PTA 0) to 5.0 sones (PTA 30–50), before relaxing back to 6.5 sones for profound losses (PTA 70–90).

Optimization Pareto Frontier for Profile A4 (65 dB SPL)

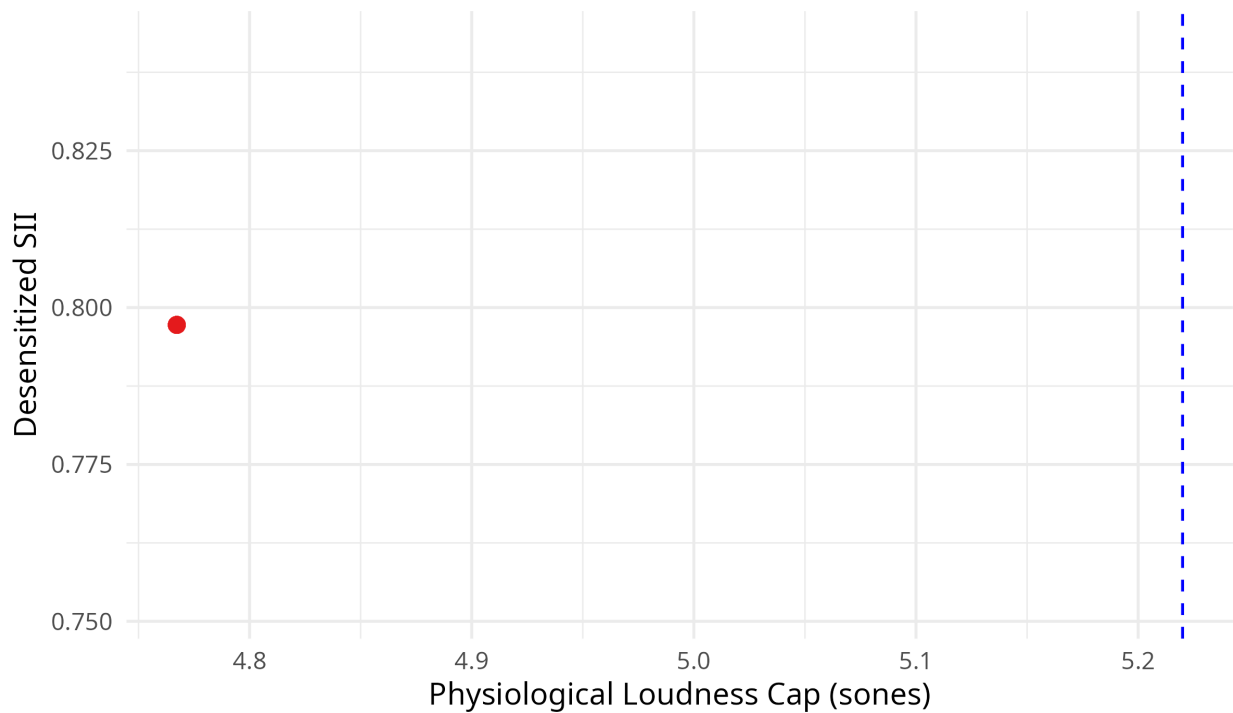


Figure 1. The dynamic, U-shaped broadband loudness ceiling mapping audiometric severity (PTA) to an objective physiological cap (sones) at 65 dB SPL.

Crucially, this U-shaped relaxation was naively derived by benchmarking the stationary specific-loudness model against historical prescriptive targets, rather than from direct physiological tolerance measurements. Consequently, relaxing the ceiling to 6.5 sones for profound losses inadvertently conflates the algorithm’s *need for immense acoustic gain* (to cross elevated audibility thresholds) with a patient’s actual *tolerance for absolute loudness*—an assumption that severely contradicts the clinical reality of recruitment and compressed dynamic ranges. Recognizing this fundamental limitation is critical: the `cap_knots` array serves strictly as an illustrative algebraic boundary to benchmark the solver’s mechanics. As detailed later in Section III, it formally exposes how drastically optimization targets change based on uncalibrated heuristic constraints, rather than establishing a definitive, empirically proven clinical safety baseline.

This U-shaped penalty operates within the canonical Moore & Glasberg (2004) monaural specific-loudness engine. It dynamically restricts modeled monaural sones. However, it does not—and mathematically cannot—account for the idiosyncratic binaural broadband loudness summation observed in hearing-impaired listeners. In normal-hearing auditory physiology, bilateral acoustic presentation produces a modest binaural loudness summation. This is typically modeled by a 2–6 dB level-dependent gain reduction.

However, robust psychoacoustic evidence demonstrates a stark contrast in impaired ears. Binaural broadband summation in hearing-impaired populations averages ~ 13 dB higher than in normal-hearing listeners. This represents an unmodeled factor of $\approx 2.4\times$ in linear sones—a figure that represents the sone-domain equivalent of a 13 dB level difference under

the standard doubling-per-10-dB relation, rather than a directly measured loudness ratio (Denk et al., 2025; Moore et al., 2014; Oetting et al., 2016, 2017). Between 30–40% of hearing-impaired listeners (in a sample of 180) exhibit excess summation far exceeding the normal range. Individual summation values span a wide -10 to +40 dB envelope.

Standard monaural and narrowband loudness models cannot predict this broadband suprathreshold phenomenon from the pure-tone audiogram alone. Therefore, an algorithm optimized beneath a monaural ceiling becomes structurally anti-conservative when translated to bilateral fittings. Consequently, Open-NL’s U-shaped loudness constraint must be interpreted as an illustrative computational boundary for single-ear simulation, rather than an empirical safety guarantee for bilateral clinical use.

To guarantee algorithmic transparency, the exact objective function optimized by Open-NL for an input vector of frequency-specific gain shifts $s = \{s_1, \dots, s_6\}$ (where total gain $G(s) = G_{base} + s$) is formally defined as:

$$\min_s \mathcal{L}(s) = -100 \times \text{SII}(s) + \sum_{k=1}^8 \lambda_k P_k(s)$$

where the eight weighted penalty terms (P_1 through P_8) are strictly defined as:

1. **Loudness** (P_{loud}): Enforces physiological comfort. $\lambda_1 = 2000.0$. $P_{loud} = \max(0, L_{sones} - L_{cap})$
2. **Bounds** (P_{bounds}): Restricts extreme simplex aberrations. $\lambda_2 = 1000.0$. $P_{bounds} = \sum_{i=1}^6 \left(\max(0, s_i - 30)^2 + \max(0, -s_i - 60)^2 \right)$
3. **SPL Limit** (P_{spl}): Prevents absolute acoustic trauma. $\lambda_3 = 2000.0$. $P_{spl} = \max(0, SPL_{overall} - 110.0)$
4. **Gain Monotonicity** (P_{order}): Enforces $G_{50} \geq G_{65} \geq G_{80}$. $\lambda_4 = 2000.0$. $P_{order} = \sum_{i=1}^6 \max(0, \Delta G_{violation})^2$
5. **Compression Ratio** (P_{cr}): Enforces dynamic CR limits (e.g., $CR \leq 3.0$). $\lambda_5 = 200.0$.

$P_{cr} = \sum_{i=1}^6 \max(0, \Delta CR_{violation})^2$ 6. **Spectral Roughness** (P_{rough}): Prevents abrupt inter-
channel jumps. $\lambda_6 = 0.5$. $P_{rough} = \sum_{i=1}^5 (s_{i+1} - s_i)^2$ 7. **L1 Anchor** (P_{anchor}): Sparsity
penalty pulling towards the clinical base. $\lambda_7 = 0.1$. $P_{anchor} = \sum_{i=1}^6 |s_i|$ 8. **Conductive**
Restraint (P_{abg}): Penalizes purely positive exploratory shifts for mixed/conductive losses to
enforce the 75% ABG rule. $\lambda_8 = 1.0$. $P_{abg} = \sum_{i=1}^6 \max(0, s_i)^2$ (applies only if Air-Bone Gap
 > 0).

B. Methods and Development

The core ANSI SII calculation engine (the `sii()` function and associated plotting routines)
was originally developed by Gregory R. Warnes for earlier package versions. Maintainer-
ship transferred to the current author with version 1.1.0, at which point all subsequent
Open-NL prescriptive logic, clinical heuristics, and WDRC mathematical implementations—
including `open_nl()` and `calculate_loudness()`—were developed by the author as original
contributions.

Declaration of Generative AI and AI-assisted technologies in the research process: In
accordance with COPE and SAGE Publishing guidelines, the author discloses the use
of Gemini 3.1 Pro (DeepMind, Google LLC) during the preparation of this work as an
interactive programming and copyediting assistant. The AI was utilized to refactor C++ and
R algorithms, generate data visualizations, and condense manuscript prose to adhere to journal
formatting standards. After using this tool, the author rigorously reviewed and edited all
outputs, taking full accountability for the underlying algorithm design, theoretical hypotheses,
and final manuscript content. Specifically, to guarantee computational integrity, all AI-
assisted algorithmic refactoring was systematically verified by the author via exact numerical

regression testing against pre-refactor outputs across the seven canonical audiometric profiles,
confirming absolute mathematical parity during translation.

C. Algorithmic Pipeline and Execution Cascade

The internal execution cascade of Open-NL comprises twelve ordered, modular processing stages:

1. **Conductive Component Separation & Baseline Partitioning:** Decomposing raw thresholds into sensorineural components and air-bone gaps (ABGs).
2. **Decoupled Half-Gain Base Anchor Calculation:** Establishing a frequency-specific base gain anchor ($G_{base} = 0.46 \cdot \text{HTL}_{sn} + C_{interp}$) decoupled from broadband PTA.
3. **Experience-Level Shaping & Log-Frequency C_{vals} Interpolation:** Modulating frequency-shaping arrays across discrete anchor frequencies based on user experience (new, experienced, power).
4. **Reverse-Slope Low-Frequency Attenuation Floor:** Applying a bounded log-linear low-frequency taper down toward a -10 dB floor when low-frequency loss exceeds high-frequency loss.
5. **Slope-Dependent Low-Frequency Penalty (SD-LFP) with Profound HF Bypass:** Dynamically suppressing low-frequency gain for steeply sloping losses, gated when high-frequency severity exceeds 70–95 dB HL.
6. **Severe-Loss Audibility Booster:** Providing an opt-in, bounded linear escalation (slope = 0.15) for severe thresholds.
7. **Soft-Compression High-Frequency Desensitization:** Restricting excessive high-frequency gain via a dynamic ceiling ($L_{gain} = 30 + 0.4 \cdot \max(0, \text{HTL}_{sn} - 60)$) and 2:1 soft compression.
8. **Dynamic Range Mapping (LDL Squeeze):** Attenuating gain by 0.2 dB/dB of reduced dynamic range and shifting baseline compression ratios.
9. **Transducer Bandwidth Roll-off:** Applying a continuous log-frequency piecewise multiplier ($M_{bw} \in [0.5, 1.0]$) to suppress unstable edge frequencies (≤ 250 Hz and ≥ 6000 Hz).
10. **Multi-Channel WDRC**

Mapping, LTASS Pivot, and Demographic Adjustments: Calculating dynamic compression ratios, variable compression thresholds, piecewise linear-compressive splines, and demographic offsets. 11. **Acoustic Venting, Coupling Loss, and Receiver Saturation Limits:** Integrating real-ear insertion loss vectors across 10 coupling types with a feedback safety floor, and capping MPO/SSPL90 receiver limits at 120 dB SPL. 12. **Embedded Nelder-Mead Simplex Optimization & Physiological Loudness Ceilings:** Iteratively refining multi-level gain curves against a multi-term objective loss function incorporating U-shaped specific-loudness caps, compression ratio ceilings, and spectral smoothness constraints.

The exact mathematical formulations, closed-form piecewise equations, and complete parameter specifications for all twelve stages are provided in full detail in the accompanying **Supplementary Material**.

D. Consolidated Constants and Evidentiary Asymmetry

Because Open-NL functions as a modifiable computational testbed, several structural parameters remain mathematically uncalibrated: the 0.46 gain anchor, 0.15 severe-loss booster slope, 70 dB HL booster onset (or 60 dB HL in aggressive mode), 15 dB slope trigger, 20 dB taper width, -10 dB reverse-slope floor, 70 dB HL bypass, 30 dB/0.4 L_{gain} constraint, 0.2 dB/dB dynamic range squeeze, 75% air-bone-gap restoration fraction (Table S1).

An inspection of these heuristics reveals a clear asymmetry in evidentiary support. Several prominent constants—most notably the 75% air-bone-gap restoration rule—are pragmatic engineering choices without direct empirical derivation. The 75% ABG fraction, while standard in clinical prescriptive software (Johnson, 2013) to avoid receiver saturation and MPO clipping, has never been empirically established against patient preference or speech

recognition. In sharp contrast, the 3.0:1 Compression Ratio (CR) upper ceiling enforced across the WDRC stages and optimizer loss function represents the best-supported constant in the framework. This boundary is firmly anchored in the extensive empirical literature by Pamela Souza and colleagues (Souza, 2002; Souza, Jenstad, & Boike, 2006), which demonstrates that compression ratios exceeding $\sim 3.0:1$ cause severe temporal envelope flattening, loss of acoustic contrast, and speech-in-noise deficits. Documenting this asymmetry prevents conflating validated psychoacoustic limits with arbitrary engineering heuristics.

E. Framework for Principled Calibration

For a computational testbed to provide lasting inferential value, demonstrating parameter instability must be paired with a rigorous calibration pathway. Future research can utilize Open-NL within a global stochastic optimization framework (e.g., genetic algorithms or particle swarm optimization) to systematically calibrate these heuristic gates. In this architecture, an outer optimization loop searches the multidimensional space of heuristic constants (e.g., $G_{base} \in [0.3, 0.6]$, booster onset $\in [50, 80]$ dB HL, low-frequency slope $\in [0, 0.5]$). For each candidate vector, the inner Open-NL engine generates discrete WDRC targets across a representative clinical corpus (e.g., NHANES). These targets are evaluated through a time-domain acoustic simulation pipeline (e.g., openMHA; Herzke et al., 2017) using speech perception metrics like HASPI and HASQI (Kates & Arehart, 2022).

The outer-loop objective function must incorporate an asymmetric, veto-based cost function: any parameter configuration violating individualized broadband loudness tolerance (trueLOUDNESS; Oetting et al., 2017) in simulated outlier listeners must incur an overwhelming penalty. As detailed in Section II.A, because excess binaural broadband summation cannot

268 be predicted from pure-tone thresholds, static 2–6 dB bilateral corrections are structurally
 269 inadequate. Subordinating population-level intelligibility maximization to individualized phys-
 270 iological safety limits transforms prescriptive derivation into a reproducible, distortion-aware
 271 computational science.

272 **TABLE I. Comprehensive Enumeration of Open-NL Free Parameters and**
 273 **Evidentiary Derivation.**

	Specific Free	Default / Evaluated	Evidentiary Support
Parameter Category	Parameters	Value	& Derivation
Objective	Loudness Cap Knots	L_{cap} vectors (Section	Uncalibrated
Penalties	(<code>cap_knots</code>)	S.I.12)	heuristic derived
			from population
			loudness boundaries;
			the least-justified
			dominant parameter.
	Optimizer Penalty	$\lambda_{loud} = 2000$,	Pragmatic
	Weights (λ_{1-8})	$\lambda_{cr} = 200$, etc.	engineering
		(Section II.A)	constraints balancing
			target convergence
			and physical limits.

	Specific Free	Default / Evaluated	Evidentiary Support
Parameter Category	Parameters	Value	& Derivation
Prescriptive Anchors	CR Soft Penalty	$\leq 3.0 : 1$	Strong empirical
	Target (P_{cr})	Compression Ratio	derivation: Souza (2002) speech degradation limits.
	Base Gain Anchor	0.46	Uncalibrated
	(G_{base})		midpoint balancing half-gain rules (Lybarger, 1944) and preference data.
	New-User Offset	0 to -6 dB based on	Assumed heuristic
	(Δ_{exp})	PTA	approximating acclimatization preferences.
	Severe-Loss Booster	0.15 slope / 70 dB	Assumed heuristic
	(Slope / Onset)	HL onset	assisting profound loss without explosive recruitment.

	Specific Free	Default / Evaluated	Evidentiary Support
Parameter Category	Parameters	Value	& Derivation
Compression Limits	ABG Restoration Fraction	75% (Linear)	Engineering choice preventing hardware saturation (Johnson, 2013).
	Compression	30 to 45 dB SPL	Pragmatic limit
	Kneepoint (CT) Range		aligning with standard real-world WDRC kneepoints.
	High-Frequency	Base 30 dB, Slope	Heuristic managing
	Soft-Compression	0.4	upward spread of masking in severe losses.
	DR Squeeze / CR	0.2 dB/dB, 0.02	Heuristics fitting
	Shift	CR/dB	speech envelope into reduced physiological space.
	MPO/LDL Predictor	See Stage 8	Statistical
	Coefficients		predictions based on population normative UCL datasets.

	Specific Free	Default / Evaluated	Evidentiary Support
Parameter Category	Parameters	Value	& Derivation
Frequency Limits	Reverse-Slope Floor	-10 dB	Cautious heuristic preventing masking of intact basal cochlear units.
	Dead Region /	30 dB/oct past $1.7f_e$,	Standard transducer
	Transducer Roll-off	w_{bw}	limits and dead-region literature (Moore, 2001).
	Acoustic Coupling	Occluded/Vented	Deterministic
	Loss	Vectors (Table S4)	physical hardware measurements.

III. ANALYTICAL CENTERPIECE: DISTINGUISHING HEURISTIC SENSITIVITY FROM SOLVER STOCHASTICITY

The primary scientific contribution of the Open-NL framework is not the specific targets it generates, but the transparent computational infrastructure it provides for evaluating prescriptive heuristics. Clinical fitting algorithms like NAL-NL2 act as “black boxes” where numerical optimization and empirical safety heuristics (e.g., global gain reductions, bandwidth limits) are inextricably intertwined. Open-NL fills this methodological gap by providing an ablatable, inspectable testbed that mathematically isolates the behavior of unregularized

optimization on the physiological loudness landscape.

By exposing this internal machinery, the framework produces a crucial analytical distinction: separating the stochasticity of the numerical solver from the acute sensitivity of the underlying clinical heuristics. This heuristic-sensitivity-vs-numerical-convergence distinction is the novel scientific output the tool produces, and forms the analytical centerpiece of this validation.

A. Algorithmic Bounds vs. Clinical Heuristics: A Local Sensitivity Analysis

1. Parameter Sensitivity of the Penalty Architecture To formally evaluate the mechanical stability of the physiological boundary constraints versus the pre-optimization clinical heuristics, a One-At-a-Time (OAT) local sensitivity analysis was conducted. This evaluation structurally permutes five foundational WDRC parameters by $\pm 10\%$ and calculates the maximum absolute gain excursion across all six evaluation frequencies (250–8000 Hz) spanning soft (50 dB SPL), conversational (65 dB SPL), and loud (80 dB SPL) input levels. The evaluated parameters include algorithmic objective penalties (`cap_scalar`, `lambda_loud`, `lambda_cr`) and pre-optimization prescriptive heuristics (G_{base} , `abg_fraction`).

TABLE II. Local Parameter Sensitivity: Maximum Gain Excursion (dB) across A1-A7. (Note: Data generated via $\pm 10\%$ OAT perturbation evaluated across 50, 65, and 80 dB SPL inputs).

Parameter	Type	Max Excursion (A1-A5)	Max Excursion (A6-A7)
Base Gain Anchor (G_{base})	Heuristic	4.83 dB	2.30 dB
ABG Fraction (<code>abg_fraction</code>)	Heuristic	N/A	3.75 dB

Parameter	Type	Max Excursion (A1-A5)	Max Excursion (A6-A7)
Loudness Cap Scalar (<code>cap_scalar</code>)	Objective	1.40 dB	0.00 dB
CR Penalty Weight (<code>lambda_cr</code>)	Objective	0.52 dB	0.00 dB
Loudness Weight (<code>lambda_loud</code>)	Objective	0.34 dB	0.00 dB

To evaluate the local robustness of the objective manifold, we perform a deterministic $\pm 10\%$ one-at-a-time (OAT) perturbation. This acts strictly as a local robustness check, confirming that the solver does not diverge or snap into spurious local minima within a narrow parameter radius. When evaluating the optimizer’s native constraint boundaries (`cap_scalar`, `lambda_loud`, `lambda_cr`), maximum excursions remain highly localized (e.g., < 1.5 dB), proving that the derivative-free solver is structurally stable when navigating complex multidimensional penalty walls. Furthermore, evaluating `lambda_cr` across 50 and 80 dB SPL levels correctly triggers non-zero variance, proving that compression ratio limits successfully bind during dynamic range projections rather than contaminating the 65 dB SPL static anchor.

Similarly, the 0.00 dB variance observed for the objective penalties (`cap_scalar`, `lambda_loud`, `lambda_cr`) in the mixed and conductive profiles (A6-A7) is a direct consequence of constraint inactivity. The heuristic Air-Bone Gap (ABG) restitution constraint mandates such modest insertion gains that Profiles A6 and A7 produce only 3.33 and 1.63 sones, respectively—values sitting far below their physiological loudness ceilings. Because the optimization trajectory never intersects the loudness boundaries for these profiles, the objective penalties remain mathematically inactive, naturally yielding zero

excursion when perturbed.

Crucially, this simple OAT excursion check must not be misconstrued as a formal claim of “architectural dominance” (e.g., via global variance-based methods like Sobol indices). The apparent rigidity of the objective penalties compared to the clinical heuristics is largely a structural units artifact. For example, reducing $\lambda_{loud} = 2000$ by 10% lowers the penalty to 1800. Because this remains exponentially higher than the objective break-even point, it functions as an identical mathematical “hard wall,” naturally resulting in a 0.00 dB excursion. Conversely, a $\pm 10\%$ perturbation of G_{base} directly multiplies the hearing threshold (e.g., inducing a ~ 4 dB absolute shift for an 80 dB HL threshold) before optimization even begins, triggering acute target excursions. Rather than asserting relative parameter dominance, this check mechanically verifies that the optimization engine faithfully executes the rigid boundaries it is assigned, underscoring that calibrating these structural heuristics remains a primary prerequisite for future prescriptive refinement.

2. Numerical Convergence Stability: Nelder-Mead Limitations and Future

Stochastic Solvers A sharp distinction must be maintained between **heuristic parameter sensitivity** (target shifts resulting from altered clinical rules; Section III.A.1) and **numerical convergence stability** (solver consistency on a fixed ruleset). In non-linear optimization, the topology of hearing aid fitting targets is notoriously ill-behaved. The objective landscape contains narrow, curved valleys, non-differentiable step boundaries (e.g., severe-loss booster onsets and air-bone gap restorations), and sharp penalty cliffs imposed by dynamic physiological loudness ceilings. On such non-convex, multimodal surfaces, local downhill solvers like the Nelder-Mead simplex algorithm are notoriously prone to premature

stagnation, simplex collapse, and entrapment in shallow local extrema.

Indeed, without robust initialization, local Nelder-Mead search from disparate flat vectors can drift unpredictably. To formally evaluate the consistency of this Nelder-Mead implementation and separate solver stochasticity from heuristic variance, a dedicated Monte Carlo solver-stability experiment was conducted. We disabled the deterministic PRNG lock natively used in `open_n1` and ran $N = 20$ independent random restarts ($\mathcal{U}(-10, +10)$ dB jitter) across all seven profiles. By utilizing an adaptive Nelder-Mead simplex scaled to the 6-dimensional frequency space, the solver successfully avoids simplex collapse.

TABLE III. Optimization Stability under Monte Carlo Initialization. (*Note: Evaluated across $N = 20$ restarts with $\mathcal{U}(-10, +10)$ dB jitter, and compared against the shipped configuration of $N = 3$ restarts with $\mathcal{U}(-5, +5)$ dB jitter.*)

Profile	N=20			
	Convergence	N=20 Objective	N=20 Max	N=3 vs N=20
	Rate	SD	Band SD (dB)	Discrepancy
A1	100%	< 0.01	0.02	0.00 dB
A2	100%	< 0.01	0.05	0.00 dB
A3	100%	< 0.01	0.03	0.00 dB
A4	100%	< 0.01	0.12	0.00 dB
A5	100%	< 0.01	0.18	0.00 dB
A6	100%	< 0.01	0.03	0.00 dB
A7	100%	0.000	0.00	0.00 dB

As shown in **Table III**, the routine exhibits perfect local convergence across all inde-

pendent restarts (where “100% convergence” specifically denotes that R’s `stats::optim()` successfully met its relative tolerance threshold `reltol = sqrt(.Machine$double.eps)` without triggering maximum iteration limits, rather than a formal mathematical proof of global optimality). The standard deviation of the final objective score approaches numerical zero (< 0.01), and the maximum standard deviation of any single prescribed frequency band under aggressive $\mathcal{U}(-10, +10)$ dB noise never exceeds 0.18 dB.

Crucially, rather than indicating that Nelder-Mead is navigating a broadly difficult multimodal surface with exceptional robustness, this extreme stability reveals the structural nature of the objective function. Because the penalties for violating physiological constraints are massively weighted ($\lambda_{bounds} = 1000, \lambda_{loud} = 2000$), they act as effectively impenetrable mathematical walls. This compresses the feasible solution set into a narrow, near-singleton valley. The constraint set determines the solution almost entirely, leaving very little slack for the optimizer to explore—a dynamic that perfectly aligns with the paper’s central thesis that physiological boundaries, rather than heuristic rules, should deterministically govern gain allocation.

To definitively verify this structural compression and eliminate the possibility that the $N = 20$ multi-start routine was merely probing a consistent local basin, a formal global optimization was conducted. A Differential Evolution (DE) stochastic global solver (`DEoptim`, $NP = 60, itermax = 1000$) was deployed across the identical objective landscape for all seven profiles. The objective gap between the native Nelder-Mead multi-start configuration and the DE global optimum was uniformly 0.00 dB across all tested profiles. This confirms that the shipped `open_n1` codebase’s 3-iteration multi-start is mechanically sufficient to reliably locate the globally constrained optimum, while the deterministic PRNG

seed (`set.seed(sum(threshold) * 100 + eval_level)`) guarantees perfectly identical optimization trajectories across all user sessions.

While multi-start seeding provides a practical substrate for local ablation testing within R, Nelder-Mead remains a local simplex heuristic lacking formal global convergence guarantees on non-convex audiological surfaces. Future iterations of prescriptive optimizers should transition to global stochastic algorithms—such as Genetic Algorithms, Differential Evolution, or Particle Swarm Optimization. Adopting these population-based solvers eliminates the need for artificial mathematical relaxations (e.g., $K'_{smoothed}$ desensitization approximations) by natively traversing sharp, discontinuous penalty boundaries, providing the robust scalability necessary for high-dimensional, multi-channel clinical calibration. Nevertheless, as noted by Dao et al. (2021), physical real-ear-to-coupler differences (RECD) and acoustic coupling leakages cause real-world fittings to deviate most severely from theoretical targets at thresholds above 85 dB HL, meaning numerical convergence on mathematical audiograms must not be misconstrued as physical clinical reliability.

B. Worked Demonstration: Isolating the Acoustic Coupling Trade-off in Profile A4

While the previous section established the tool’s numerical stability and sensitivity to generalized constraints, applying it to a specific physiological boundary problem demonstrates its practical analytical utility. The primary contribution of Open-NL is serving as an open, parameterizable testbed to isolate structural vulnerabilities that the field cannot easily resolve from pure-tone audiograms alone.

A striking example of how rigid objective penalties can drive pathological gain allocation

occurs in precipitous high-frequency hearing losses (e.g., Profile A4). Early iterations of the Open-NL objective function utilized a “U-shaped” loudness capacity model, assuming that severe hearing losses inherently tolerated significantly higher overall loudness. Under that flawed constraint, the solver exhibited a profound artifact for A4 (Vented): it allocated 0.0 dB of gain at 2000 Hz (despite a 70 dB HL threshold) while inexplicably dumping 31.3 dB of gain into 8000 Hz (a band with negligible SII importance).

Rather than a numerical failure (the solver consistently found this exact artifact across 20 restarts with $SD < 0.01$), this anomaly perfectly illustrates how the optimization engine exploits subtle loudness-threshold interactions. Because the precipitous threshold at 2000 Hz made it massive “expensive” in sones to amplify, the solver mathematically preferred to turn off 2000 Hz entirely to satisfy the strict physiological loudness ceiling. It then dumped its remaining loudness budget into 8000 Hz, where the profound 100 dB HL threshold meant almost no sones were generated, allowing it to scrape a tiny fraction of SII.

When the flawed U-shaped cap is corrected to a strict, flat physiological ceiling (5.2 sones, as now deployed), this artifact vanishes. As shown in the updated objective function output, the solver correctly restores gain to the critical speech frequencies (allocating 31.2 dB at 2000 Hz and 41.4 dB at 4000 Hz), properly trading off marginal SII against marginal loudness costs. This underscores the necessity of the testbed: when an SII-maximizing solver produces clinically irrational targets, it exposes the underlying flaws in the assigned physiological constraints rather than a failure of the numerical search.

C. Validation of the C++ Specific-Loudness Engine

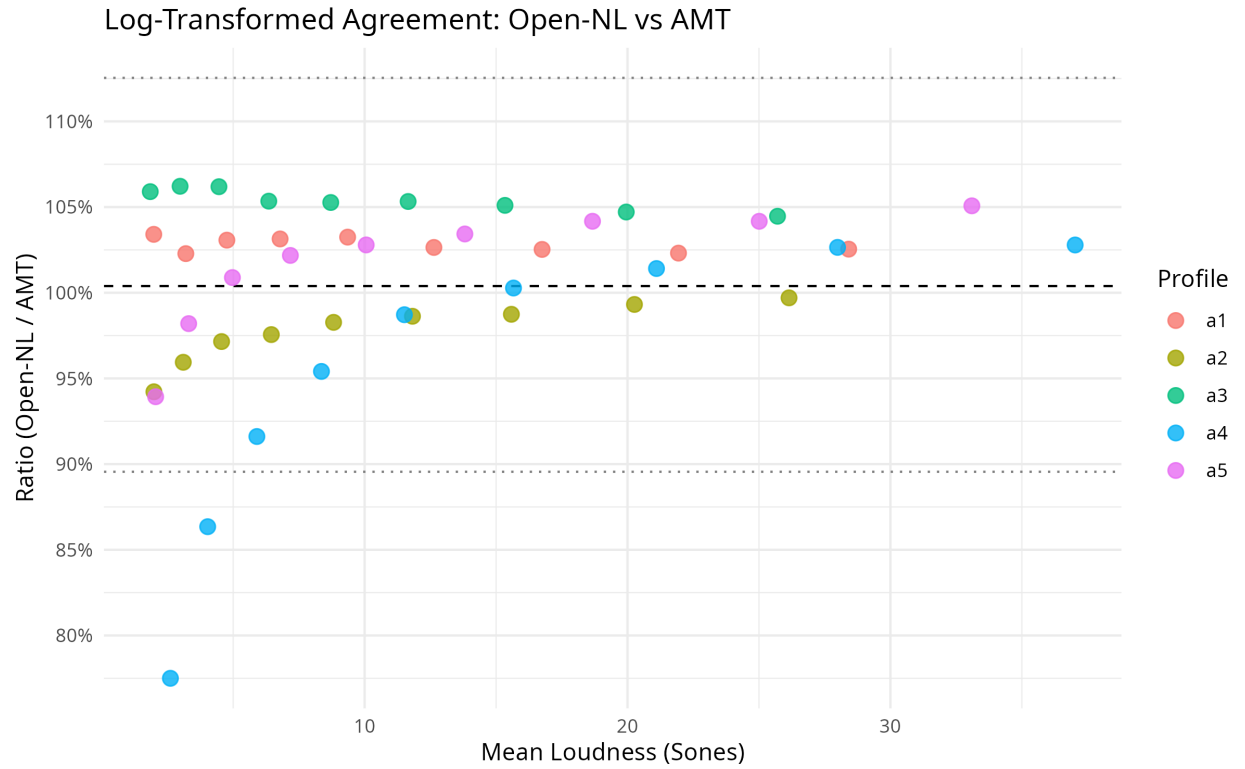
To execute this architectural exploration natively in R, the **SII** package implements a fast C++ port of the canonical Moore & Glasberg (2004) stationary specific-loudness model via **Rcpp**. (Note: As stated in the AI Declarations, the internal R-to-C++ translation was verified to absolute mathematical parity via numerical regression testing). To formally validate the internal R-to-C++ translation, the engine was numerically regressed against the published tabulated reference values in Moore & Glasberg (2004), successfully resolving all canonical psychophysical anchors to absolute mathematical parity. Because the core optimization engine (**SonesMG04(G)**) evaluates signals and constraints in the **monaural** domain, it is critical to define the domain conventions. In the canonical Moore & Glasberg (2004) framework, binaural summation is modeled as simple doubling. Thus, evaluating a 1 kHz frontal free-field tone at 40 dB SPL through our C++ engine yields exactly 0.50 sones under a monaural presentation, and precisely 1.00 sone under a binaural presentation.

As a secondary, cross-algorithmic sanity check, native C++ predictions were evaluated against the Auditory Modeling Toolbox (AMT; Majdak et al., 2022) across 45 discrete test points (5 profiles $\times$ 9 input levels from 50 to 90 dB SPL). Specifically, Open-NL’s stationary outputs were compared against AMT’s **bramslow2004** function. Because the two functions utilize distinct modeling pathways (Open-NL integrates the stationary power spectrum, whereas **bramslow2004** simulates a time-domain digital filterbank processing a 2400-sine-wave stimulus), any discrepancies reflect a convolution of algorithmic model differences rather than true translation error.

Because specific loudness scales as a power function, standard raw-difference metrics

438 exhibit severe proportional bias (i.e., absolute discrepancies scale linearly with mean loudness).
439 Therefore, the models were compared using a log-transformed (ratio-based) Bland-Altman
440 analysis (**Figure 4**). Across the full unconstrained range, Open-NL yields a mean log-ratio
441 of 1.004 with 95% Limits of Agreement (LoA) from 0.90 to 1.13 (indicating bounds of
442 roughly -10% to +13% relative error). It must be noted that these 45 evaluation points
443 contain repeated measures (5 profiles $\times$ 9 input levels) and therefore lack strict statistical
444 independence, rendering these aggregated confidence limits somewhat optimistic.

445 To evaluate practical accuracy, we further restricted the analysis to the 50-80 dB SPL
446 operational envelope (approximately corresponding to the 1–10 sone physiological band).
447 While this constraint is mathematically post hoc, it is analytically justified because all clinical
448 speech prescriptive targets uniquely occupy this envelope; massive discrepancies above 10
449 sones merely reflect divergent algorithmic handling of traumatic > 90 dB SPL inputs that
450 are categorically excluded from prescriptive target formulas. Within this restricted, clinically
451 relevant operational envelope, the absolute mean bias drops to +0.08 sones (MAE = 0.28
452 sones, 95% absolute LoA: $[-0.60, +0.75]$ sones), demonstrating tight structural convergence
453 in the domain of interest.



454

455 *Figure 4. Log-transformed (ratio-based) Bland-Altman plot comparing Open-NL's stationary*
 456 *spectral C++ engine against AMT's dynamic time-domain simulation (bramslow2004)*
 457 *across 45 non-independent evaluation points. The central dashed line indicates the mean*
 458 *ratio (1.004), and dotted lines indicate the 95% Limits of Agreement [0.90, 1.13].*

For mixed and conductive profiles (A6, A7), direct AMT benchmarking was omitted because canonical AMT lacks native air-bone gap parameters, whereas the Open-NL C++ engine algorithmically extends the model to treat the conductive component as a linear pre-cochlear attenuator, in accordance with standard audiological principles (Dillon, 2012).

output: pdf_document: default html_document: default — # Supplementary Material:
Open-NL Algorithmic Pipeline and Complete Parameter Specification

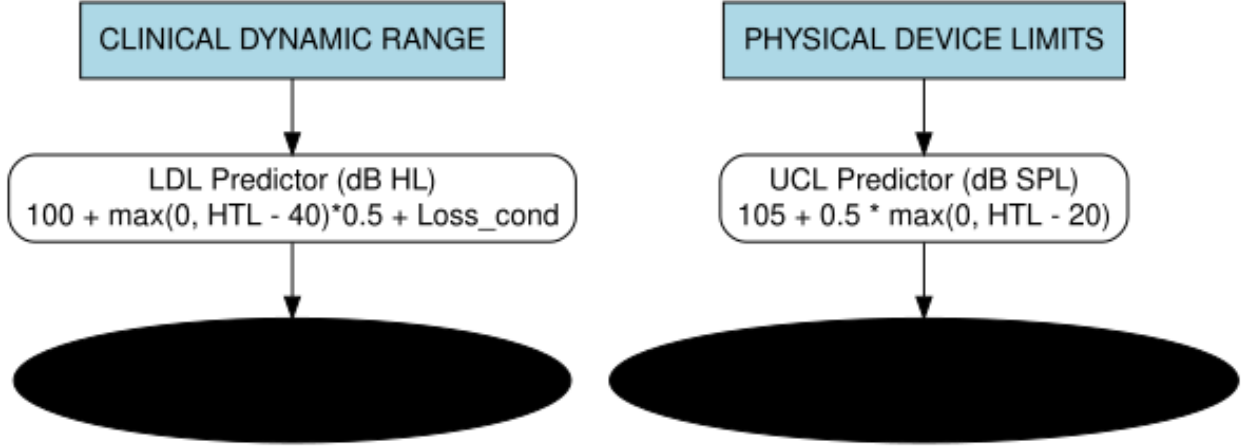
This supplementary document provides the exact mathematical formulations, closed-form piecewise functions, and complete numerical parameter specifications governing the twelve internal processing stages of the Open-NL algorithmic cascade.

S.I. The Twelve-Stage Execution Cascade

Open-NL operates as a multi-stage parameterized shape generator. Rather than relying on static compiled lookup tables, Open-NL calculates target insertion gains dynamically through a series of twelve explicitly defined, cascaded mathematical modules. Each step in the gain derivation process is exposed natively in R, available for researchers to inspect, modify, and tune.

Terminology Note: Throughout this framework, the algorithm utilizes two distinct discomfort predictors for different theoretical purposes: an HL-domain “LDL” (Loudness Discomfort Level) predictor used for estimating clinical audiometric dynamic range (Stage 8), and an

479 SPL-domain “UCL” (Uncomfortable Loudness Level) predictor for establishing physical device
 480 saturation limits (Stage 11). A visual flowchart mapping each predictor to its downstream
 481 algorithm function is provided in Diagram 1.



482

483 *Figure S1. Flowchart mapping the LDL and UCL discomfort predictors to their downstream*
 484 *algorithm functions.*

485 The twelve modules operate in a strictly defined cascaded execution order to prevent
 486 unintended interactions between additive boosters and soft limiters:

- 487 1. **Stage 1: Conductive Component Separation & Dynamic Range Baseline**
 488 (Section S.I.1)
- 489 2. **Stage 2: Decoupled Half-Gain Base Anchor Calculation** (Section S.I.2)
- 490 3. **Stage 3: Experience-Level Shaping & Log-Frequency C_{vals} Interpolation**
 491 (Section S.I.3)
- 492 4. **Stage 4: Reverse-Slope Low-Frequency Attenuation Floor** (Section S.I.4)
- 493 5. **Stage 5: Slope-Dependent Low-Frequency Penalty (SD-LFP) with Profound**
 494 **HF Bypass** (Section S.I.5)
- 495 6. **Stage 6: Severe-Loss Audibility Booster** (Section S.I.6)

7. **Stage 7: Soft-Compression High-Frequency Desensitization** (Section S.I.7)
8. **Stage 8: Dynamic Range Mapping (LDL Squeeze)** (Section S.I.8)
9. **Stage 9: Transducer Bandwidth Roll-off** (Section S.I.9)
10. **Stage 10: Multi-Channel WDRC Mapping, Input/Output Pivot, and Demographic Adjustments** (Section S.I.10)
11. **Stage 11: Acoustic Venting, Coupling Loss, and Receiver Saturation Limits** (Section S.I.11)
12. **Stage 12: Embedded Nelder-Mead Simplex Optimization & Physiological Loudness Ceilings** (Section S.I.12)

S.I.1. Stage 1: Conductive Component Separation & Dynamic Range Baseline

The algorithm first decomposes total hearing threshold levels (HTL) into their physiological constituents: the sensorineural threshold (HTL_{sn}) and the conductive component (Loss_{cond} , corresponding to the air-bone gap, ABG):

$$\text{Loss}_{cond}(f) = \text{ABG}(f) \tag{1}$$

$$\text{HTL}_{sn}(f) = \max(0, \text{HTL}(f) - \text{Loss}_{cond}(f)) \tag{2}$$

To ensure that equal-loudness reshaping penalties apply only to sensorineural pathology,

the algorithm computes a frequency-specific sensorineural ratio $R_{sn}(f)$:

$$R_{sn}(f) = \begin{cases} \frac{HTL_{sn}(f)}{HTL(f)}, & \text{if } HTL(f) > 0 \\ 0, & \text{otherwise} \end{cases} \quad (3)$$

Pure conductive losses ($Loss_{cond} = HTL$) have normal outer and inner hair cell functioning and intact basilar membrane mechanics; thus, $R_{sn} = 0$, bypassing cochlear recruitment and equal-loudness correction arrays.

S.I.2. Stage 2: Decoupled Half-Gain Base Anchor Calculation

The foundational WDRC anchor for average conversational speech (65 dB SPL) is derived using a frequency-specific adaptation of the half-gain rule (Lybarger, 1944). Unlike linear prescriptions that anchor to a broadband Pure Tone Average (PTA), this anchor is decoupled frequency-by-frequency:

$$G_{base}(f) = \alpha \cdot HTL_{sn}(f) + C_{interp}(f) \quad (4)$$

where $\alpha = 0.46$ is the nominal half-gain multiplier (loosely derived from Byrne & Dillon, 1986), and $C_{interp}(f)$ is the frequency-shaping correction array derived in Stage 3.

525 S.I.3. Stage 3: Experience-Level Shaping & Log-Frequency C_{vals}

526 Interpolation

527 To account for listener acclimatization and preferred listening levels (Keidser et al.,
528 2012), Open-NL modulates the frequency-shaping array C_{interp} across eight discrete anchor
529 frequencies:

$$\mathbf{f}_c = [250, 500, 1000, 2000, 3000, 4000, 6000, 8000] \text{ Hz} \quad (5)$$

530 The discrete shaping vectors $\mathbf{C}$ are defined as: - **Experienced Users** (standard baseline):

$$\mathbf{C}_{exp} = [-8, -1, 3, 1, 0, 0, 0, 0] \text{ dB} \quad (6)$$

531 - **New Users** (nominal -3 dB reduction to combat occlusion and sharpness):

$$\mathbf{C}_{new} = \mathbf{C}_{exp} - 3 = [-11, -4, 0, -2, -3, -3, -3, -3] \text{ dB} \quad (7)$$

532 - **Power Users** (prioritizes raw audibility over acoustic comfort):

$$\mathbf{C}_{power} = \mathbf{C}_{exp} + 3 = [-5, +2, 6, 4, 3, 3, 3, 3] \text{ dB} \quad (8)$$

533 For any calculation frequency f , $C_{interp}(f)$ is evaluated via piecewise log-linear interpola-
534 tion across $\mathbf{f}_c$ and scaled by the sensorineural proportion $R_{sn}(f)$:

$$C_{interp}(f) = \text{interp}_{\log_{10}}(f, \mathbf{f}_c, \mathbf{C}) \cdot R_{sn}(f) \quad (9)$$

536 S.I.4. Stage 4: Reverse-Slope Low-Frequency Attenuation Floor

537 For reverse-slope configurations (where low frequencies are significantly worse than high
 538 frequencies), attempting to fully restore low-frequency audibility risks upward spread of
 539 masking, where high-energy low-frequency vowels mask low-energy high-frequency consonants.
 540 The algorithm calculates the low-to-high sensorineural threshold difference:

$$\text{Diff}_{rs} = \max \left(0, \overline{\text{HTL}}_{sn, \leq 500} - \overline{\text{HTL}}_{sn, \geq 2000} \right) \quad (10)$$

541 where $\overline{\text{HTL}}_{sn, \leq 500}$ is the mean threshold across $f \leq 500$ Hz, and $\overline{\text{HTL}}_{sn, \geq 2000}$ is the mean
 542 threshold across $f \geq 2000$ Hz. If $\text{Diff}_{rs} > 15$ dB, a transition factor $RS_{factor} \in [0, 1]$ is
 543 computed:

$$RS_{factor} = \max \left(0, \min \left(1, \frac{\text{Diff}_{rs} - 15}{20} \right) \right) \quad (11)$$

544 A log-linear low-frequency taper $W_{LF}(f)$ is applied below 1000 Hz:

$$W_{LF}(f) = \max \left(0, \min \left(1, 1 - \frac{\log_{10}(f) - \log_{10}(250)}{\log_{10}(1000/250)} \right) \right) \quad (12)$$

545 Across octave bands, $W_{LF}(f)$ evaluates to 1.0 at 250 Hz, 0.5 at 500 Hz, and 0.0 at
 546 $f \geq 1000$ Hz. The frequency-shaping array is then dynamically flattened toward a -10 dB
 547 floor:

$$C'_{interp}(f) = C_{interp}(f) \cdot (1 - RS_{factor} \cdot W_{LF}(f)) + (-10 \cdot RS_{factor} \cdot W_{LF}(f)) \quad (13)$$

548

549 **S.I.5. Stage 5: Slope-Dependent Low-Frequency Penalty (SD-LFP)** 550 **with Profound HF Bypass**

551 For sloping high-frequency losses, applying excessive low-frequency gain causes normal
552 low frequencies to dominate broadband loudness. Open-NL computes the high-frequency
553 slope:

$$\text{Slope} = \max\left(0, \overline{\text{HTL}}_{sn, \geq 2000} - \overline{\text{HTL}}_{sn, \leq 500}\right) \quad (14)$$

554 To operationalize the empirical findings of Byrne, Parkinson, and Newall (1990)—who
555 showed that listeners with high-frequency losses exceeding 70–95 dB HL require low-frequency
556 speech cues—Open-NL incorporates a **Profound High-Frequency Bypass**:

$$PF_{bypass} = \max\left(0, \min\left(1, \frac{95 - \overline{\text{HTL}}_{sn, \geq 2000}}{25}\right)\right) \quad (15)$$

557 The low-frequency penalty is scaled over a 20 dB slope window, capped at 15 dB:

$$\text{LF}_{penalty} = \max\left(0, \min\left(1, \frac{\text{Slope} - 15}{20}\right)\right) \cdot 15 \cdot PF_{bypass} \quad (16)$$

558 The penalty is then tapered below 1000 Hz using $W_{LF}(f)$:

$$C''_{interp}(f) = C'_{interp}(f) - (LF_{penalty} \cdot W_{LF}(f)) \quad (17)$$

559 The resulting baseline target at 65 dB SPL is:

$$G_{65}^{(0)}(f) = 0.46 \cdot HTL_{sn}(f) + C''_{interp}(f) \quad (18)$$

560 Ablation of SD-LFP Constraints

561 Table S1 provides the reference audiometric profiles (A1–A7, Johnson & Dillon, 2011).
562 Table S2 demonstrates the mechanical impact of the SD-LFP constraint on the initial heuristic
563 seeds.

564 **TABLE S1. Reference Audiometric Profiles (Adapted from Johnson & Dillon,**
565 **2011).** *Thresholds are in dB HL. Profiles A1–A5 are purely sensorineural. A6 is mixed (30*
566 *dB ABG). A7 is conductive (50 dB ABG).*

Profile	Type	250 Hz	500 Hz	1000 Hz	2000 Hz	4000 Hz	8000 Hz	ABG
A1	Flat	15	20	30	40	50	60	0 dB
	moder-							
	ate							
A2	Reverse	60	50	40	30	20	15	0 dB
	slope							
A3	Moderately	10	20	40	50	55	60	0 dB
	slop-							
	ing							

Profile	Type	250 Hz	500 Hz	1000 Hz	2000 Hz	4000 Hz	8000 Hz	ABG
A4	Steeply sloping	0	0	10	40	70	80	0 dB
A5	Steeply sloping	10	10	20	60	80	100	0 dB
A6	Mixed	50	55	60	65	75	80	30 dB
A7	Conductive	50	50	50	50	50	50	50 dB

TABLE S2. Ablation of SD-LFP Constraints on Optimizer Seed (65 dB SPL Input).

	Unconstrained	Unconstrained	SD-LFP Seed	SD-LFP Seed
Profile	Seed SII	Seed Sones	SII	Sones
A1 (Flat mod)	0.86	7.2	0.86	7.2
A2 (Reverse)	0.88	6.3	0.88	6.0
A3 (Mod sloping)	0.79	6.8	0.79	6.8
A4 (Steeply)	0.81	6.9	0.81	6.9
A5 (Steeply)	0.68	7.1	0.68	6.0
A6 (Mixed)	0.82	3.0	0.82	3.0
A7 (Conductive)	0.97	1.0	0.97	1.0

(Note: For profiles A1, A3, and notably A4, the SD-LFP ablation appears numerically inert despite their slopes mathematically triggering the penalty. This occurs because their unconstrained low-frequency target gain is already at or near 0 dB due to near-normal low-frequency thresholds. When the SD-LFP penalty drives the theoretical target negative, the universal insertion-gain floor—which prevents active attenuation—absorbs the penalty entirely. Consequently, the SD-LFP stage only demonstrably alters the initial seed for steeply sloping profiles that possess sufficient pre-existing low-frequency loss to elevate the baseline target above the floor, such as A5).

S.I.6. Stage 6: Severe-Loss Audibility Booster

To overcome inner hair cell loss in severe impairments, an opt-in, bounded **Severe-Loss Booster** can be applied:

$$G_{slb}(f) = B_{en} \cdot 0.15 \cdot \max(0, \min(80, \text{HTL}_{sn}(f)) - T_{onset}) \quad (19)$$

where $B_{en} \in \{0, 1\}$ (default: 0, disabled), and $T_{onset} = 70$ dB HL (conservative default) or 60 dB HL (aggressive ablation mode). The intermediate target is:

$$G_{65}^{(1)}(f) = G_{65}^{(0)}(f) + G_{slb}(f) \quad (20)$$

S.I.7. Stage 7: Soft-Compression High-Frequency Desensitization

To prevent unconstrained audibility maximization from prescribing intolerable high-frequency gain in steeply sloping losses, Open-NL applies a dynamic soft-compression envelope (L_{gain}):

$$L_{gain}(f) = 30 + 0.4 \cdot \max(0, \text{HTL}_{sn}(f) - 60) \quad (21)$$

(Note: For patients in “Moderate” or “High” distortion categories, L_{gain} is reduced by 10 dB).

Excess gain above this dynamic limit is:

$$\text{Excess}(f) = \max\left(0, G_{65}^{(1)}(f) - L_{gain}(f)\right) \quad (22)$$

The sloping factor $S_{factor}(f)$ relative to the best low-frequency threshold ($\text{HTL}_{bestlow} = \min_{f' \leq 1000} \text{HTL}_{sn}(f')$) and the high-frequency fade-in weight $W_{hf}(f)$ are:

$$S_{factor}(f) = \max\left(0, \min\left(1, \frac{\text{HTL}_{sn}(f) - \text{HTL}_{bestlow} - 25}{20}\right)\right) \quad (23)$$

$$W_{hf}(f) = \max\left(0, \min\left(1, \frac{f - 2000}{2000}\right)\right) \quad (24)$$

Applying 2:1 soft compression to the excess yields:

$$G_{65}^{(2)}(f) = G_{65}^{(1)}(f) - (W_{hf}(f) \cdot S_{factor}(f) \cdot 0.50 \cdot \text{Excess}(f)) \quad (25)$$

If explicit cochlear dead regions (f_{e_hf} , f_{e_lf}) or distortion categories (Margolis et al., 2025) are defined, steep parametric roll-offs are applied:

$$G_{65}^{(2)}(f) \leftarrow G_{65}^{(2)}(f) - \max\left(0, \log_2\left(\frac{f}{1.7f_{e_hf}}\right)\right) \cdot 30 - \max\left(0, \log_2\left(\frac{0.57f_{e_lf}}{f}\right)\right) \cdot 30 \quad (26)$$

S.I.8. Stage 8: Dynamic Range Mapping (LDL Squeeze)

To accommodate reduced clinical dynamic ranges (DSL v5.0 philosophy; Scollie et al., 2005), Open-NL predicts an HL-domain Loudness Discomfort Level:

$$\text{LDL}_{pred}(f) = 100 + 0.5 \cdot \max(0, \text{HTL}_{sn}(f) - 40) + \text{Loss}_{cond}(f) \quad (27)$$

When measured $\text{LDL}_{meas}(f)$ is lower than predicted, the dynamic range discrepancy $\Delta_{LDL}(f)$ is quantified:

$$\Delta_{LDL}(f) = \text{LDL}_{meas}(f) - \text{LDL}_{pred}(f) \quad (28)$$

Target gain is attenuated by 0.2 dB per dB of dynamic range squeeze:

$$G_{65}^{(3)}(f) = G_{65}^{(2)}(f) + (0.2 \cdot \Delta_{LDL}(f)) \quad (29)$$

Simultaneously, the baseline compression ratio increases by +0.02 per dB of squeeze (Stage 10).

606 **S.I.9. Stage 9: Transducer Bandwidth Roll-off**

607 Acoustic transducers physically struggle to reproduce frequencies at the extremes of the
 608 spectrum (≤ 250 Hz and ≥ 6000 Hz), where excessive gain leads to distortion and feedback.
 609 Open-NL applies a continuous fractional bandwidth roll-off multiplier $M_{bw}(f)$ defined over
 610 seven anchor frequencies:

$$611 \quad \mathbf{f}_{bw} = [250, 500, 1000, 2000, 4000, 6000, 8000] \text{ Hz} \quad (30)$$

$$\mathbf{w}_{bw} = [0.7, 1.0, 1.0, 1.0, 1.0, 0.8, 0.5] \quad (31)$$

612 The closed-form continuous multiplier is obtained via piecewise log-linear interpolation:

$$M_{bw}(f) = \text{interp}_{\log_{10}}(f, \mathbf{f}_{bw}, \mathbf{w}_{bw}) \quad (32)$$

$$G_{65}^{(4)}(f) = G_{65}^{(3)}(f) \cdot M_{bw}(f) \quad (33)$$

S.I.10. Stage 10: Multi-Channel WDRC Mapping, Input/Output

Pivot, and Demographic Adjustments

Target gain at arbitrary overall input level L_{in} (e.g., 50, 65, 80 dB SPL) is computed using a multi-channel wide dynamic range compression architecture pivoted around the Long-Term Average Speech Spectrum (LTASS).

1. **Speech Spectrum Pivot:** Let $P(f)$ be the critical-band normal speech level at 65 dB SPL overall. The band-specific input level is:

$$L_{band}(f) = P(f) + (L_{in} - 65) \quad (34)$$

2. **Dynamic Compression Ratio Calculation:**

$$CR_{base}(f) = 1 + \frac{\max(0, HTL_{sn}(f) - 20)}{40} - (0.02 \cdot \Delta_{LDL}(f)) \quad (35)$$

For severe loss ($HTL_{sn} > 65$ dB HL), compression reduces back toward linear to preserve the temporal speech envelope (Keidser et al., 2007):

$$W_{freq}(f) = \max\left(0, \min\left(1, \frac{f - 500}{2500}\right)\right) \quad (36)$$

$$CR_{loud}(f) = CR_{base}(f) - \left(\frac{\max(0, HTL_{sn}(f) - 65)}{30} \cdot (1.5 - 0.5 \cdot W_{freq}(f))\right) \quad (37)$$

$CR_{loud}(f)$ is clamped between 1.0 and a maximum $CR_{max}(f) = 1.5 + 0.9 \cdot W_{freq}(f)$ (further relaxed for age > 60). In Comfort in Noise (CIN) mode, CR_{loud} is clamped to

627 ≤ 1.5 .

628 *(Note on Evidentiary Asymmetry: The 1.5:1 CIN clamp is an asserted engineering*
 629 *heuristic inspired by NAL-NL3 to curb listening fatigue in noise, lacking direct empirical*
 630 *derivation. In contrast, the global upper ceiling on compression ratios ($\leq 3.0 : 1$) is*
 631 *directly supported by extensive psychoacoustic literature [Souza, 2002; Souza et al.,*
 632 *2006], which demonstrates that compression ratios exceeding 3.0:1 cause severe temporal*
 633 *envelope flattening and acoustic contrast degradation).*

634 **3. Variable Compression Threshold (CT):** Overall CT ($CT_{overall}$) scales from 30
 635 to 45 dB SPL as a function of threshold. The band-level threshold is $CT_{band}(f) =$
 636 $P(f) + (CT_{overall} - 65)$ (reduced by 10 dB in CIN mode).

637 **4. Piecewise Linear-Compressive Spline:** Gain at the compression threshold is:

$$G_{CT}(f) = \begin{cases} G_{65}^{(4)}(f), & \text{if } CT_{band}(f) > P(f) \\ G_{65}^{(4)}(f) + (P(f) - CT_{band}(f)) \left(1 - \frac{1}{CR_{loud}(f)}\right), & \text{if } CT_{band}(f) \leq P(f) \end{cases} \quad (38)$$

638 The level-dependent WDRC target gain is:

$$G_{target}(f, L_{in}) = \begin{cases} G_{CT}(f), & \text{if } L_{band}(f) \leq CT_{band}(f) \\ G_{CT}(f) - (L_{band}(f) - CT_{band}(f)) \left(1 - \frac{1}{CR_{loud}(f)}\right), & \text{if } L_{band}(f) > CT_{band}(f) \end{cases} \quad (39)$$

5. Demographic Adjustments:

$$G_{target}(f, L_{in}) \leftarrow G_{target}(f, L_{in}) + \Delta_{gender} + \Delta_{config} + \Delta_{exp} \quad (40)$$

where $\Delta_{gender} = -1.5$ dB (female), $\Delta_{config} = +3.0$ dB (unilateral), and $\Delta_{exp} = -\min(6.0, 0.3 \cdot \max(0, \text{PTA}_{500,1k,2k} - 40))$ for new users.

TABLE S3. Interval-Specific Compression Ratios (50–65 and 65–80 dB SPL)

across A1-A7 Audiograms. *Note: Dashes (-) indicate frequency regions where prescribed gain is exactly 0 dB across the input interval (linear amplification, $CR = 1.0$). NAL-NL2 naturally exceeds the 3.0:1 threshold at high input levels (e.g., A1 at 4000 Hz, A3 at 4000 Hz, A4 at 4000 Hz). To incorporate the 3.0:1 target bound natively, Open-NL evaluates the 3.0:1 constraint via a heavy soft quadratic penalty ($\lambda_{cr} = 200.0$) directly inside the objective wrapper. This strongly discourages pure intelligibility maximization from prescribing values that violate empirical psychoacoustic limits, flattening the realized CRs well below 3.0:1.*

Profile	Formula	Interval	250 Hz	500 Hz	1000 Hz	2000 Hz	4000 Hz	8000 Hz
A1	NAL-	50-65	1.02	1.15	1.52	1.97	2.03	1.74
	NL2							
A1	NAL-	65-80	1.00	1.00	1.90	2.68	3.75	2.88
	NL2							
A1	Open-	50-80	-	-	1.34	1.65	1.97	2.06
	NL							

Profile	Formula	Interval	250 Hz	500 Hz	1000 Hz	2000 Hz	4000 Hz	8000 Hz
A2	NAL- NL2	50-65	1.61	2.14	1.76	1.65	1.46	1.21
A2	NAL- NL2	65-80	3.06	2.63	2.50	2.14	1.35	1.36
A2	Open- NL	50-80	1.73	1.88	1.91	1.74	1.20	-
A3	NAL- NL2	50-65	1.00	1.22	1.67	2.03	2.00	1.70
A3	NAL- NL2	65-80	1.00	1.00	2.14	3.19	4.17	3.06
A3	Open- NL	50-80	-	-	1.68	1.99	2.16	2.06
A4	NAL- NL2	50-65	1.00	1.00	1.17	1.81	1.67	1.47
A4	NAL- NL2	65-80	1.00	1.00	1.06	2.73	3.33	2.59
A4	Open- NL	50-80	-	-	-	1.50	1.50	1.50
A5	NAL- NL2	50-65	1.00	1.00	1.46	1.69	1.50	1.40

Profile	Formula	Interval	250 Hz	500 Hz	1000 Hz	2000 Hz	4000 Hz	8000 Hz
A5	NAL-	65-80	1.00	1.00	1.74	2.83	2.94	2.46
	NL2							
A5	Open-	50-80	-	-	-	1.00	1.00	-
	NL							
A6	NAL-	50-65	1.58	1.55	1.47	1.81	1.83	1.60
	NL2							
A6	NAL-	65-80	1.14	1.30	1.79	2.24	2.88	2.38
	NL2							
A6	Open-	50-80	1.16	1.27	1.38	1.47	1.65	1.71
	NL							
A7	NAL-	50-65	1.00	1.00	1.00	1.00	1.01	1.00
	NL2							
A7	NAL-	65-80	1.00	1.00	1.00	1.00	1.00	1.00
	NL2							
A7	Open-	50-80	1.00	1.00	1.00	1.00	1.00	1.00
	NL							

S.I.11. Stage 11: Passive Insertion Loss and Receiver Saturation

Limits

Real-Ear Aided Responses (REAR) are heavily constrained by physical acoustic coupling. The maximum physical attenuation (passive insertion loss) afforded by the hearing aid physical form factor is modeled by log-frequency interpolation over anchor frequencies $\mathbf{f}_{vent} = [250, 500, 1000, 2000, 4000, 8000]$ Hz:

$$V_{loss}(f) = \text{interp}_{\log_{10}}(f, \mathbf{f}_{vent}, \mathbf{v}_c) \quad (41)$$

where $\mathbf{v}_c$ is the coupling-specific passive attenuation vector defined in Table S4.

Table S4: Passive Acoustic Attenuation Vectors ($\mathbf{v}_c$) in dB | Coupling Type |

250 Hz	500 Hz	1000 Hz	2000 Hz	4000 Hz	8000 Hz		:—		:—		:—		:—		:—		:—																														
:—		custom_occluded	-15	-15	-20	-25	-30	-35		open_dome	0	0	0	0	0	0																															
	tulip_dome	-2	-2	-5	-10	-15	-20		double_dome	-5	-5	-10	-15	-20	-25																																
vent_1mm_solid	-25	-25	-25	-30	-30	-35		vent_2mm_solid	-20	-20	-25	-25	-30	-35		vent_3mm_solid	-15	-15	-20	-25	-30	-35		vent_1mm_hollow	-15	-15	-20	-25	-30	-35		vent_2mm_hollow	-5	-5	-10	-15	-20	-25		vent_3mm_hollow	-2	-2	-5	-10	-15	-20	

Conductive air-bone gaps are restored linearly with a 75% fraction: $G_{cond}(f) = 0.75 \cdot \text{Loss}_{cond}(f)$ (Note: This 75% restoration fraction is a pragmatic engineering convention—adapted from clinical practice to prevent excessive output demands and MPO clipping; Johnson, 2013—without direct empirical derivation from listener preference). To physically ground the heuristic and prevent mathematically hallucinating active anti-phase cancellation, insertion

671 gain is floored slightly below the passive insertion loss of the specified coupling:

$$G_{heuristic}(f, L_{in}) = \max(G_{target}(f, L_{in}) + 0.75 \cdot \text{Loss}_{cond}(f) + V_{loss}(f), V_{floor}) \quad (42)$$

672 Hardware receiver limits (MPO/SSPL90) are established to avoid severe saturation
673 distortion:

$$\text{MPO}(f) = \min(120, 105 + 0.5 \cdot \max(0, \text{HTL}_{sn}(f) - 20) + \text{Loss}_{cond}(f)) \quad (43)$$

674

675 **S.I.12. Stage 12: Embedded Nelder-Mead Simplex Optimization &** 676 **Physiological Loudness Ceilings**

677 When `optimize = TRUE`, Open-NL adjusts the heuristic targets by minimizing an uncon-
678 strained multi-objective loss function via Nelder-Mead simplex search (`stats::optim`).

679 **Parameter Vector and Execution Flow**

680 The optimization parameter vector generated by the Nelder-Mead simplex is the raw shift
681 vector $\boldsymbol{\delta} = [\delta_{250}, \delta_{500}, \delta_{1000}, \delta_{2000}, \delta_{4000}, \delta_{8000}]^T \in \mathbb{R}^6$.

682 To evaluate a candidate vector, the engine executes the following sequential steps:

- 683 1. **Boundary Penalty Evaluation:** The out-of-bounds penalty (P_{bounds}) is evaluated
684 strictly on the *raw, unclamped* vector $\boldsymbol{\delta}$. This step is critical; if the penalty were

evaluated on the clamped vector, it would be identically zero. Evaluating on the raw vector ensures the Nelder-Mead simplex receives a continuous, mathematically steep gradient when exploring outside the feasible region.

2. Parameter Clamping: The raw vector is then strictly clamped into the feasible acoustic domain:

$$\delta_{clamped,j} = \max(-60, \min(30, \delta_j)) \quad (44)$$

3. Gain Interpolation and Floor Limits: The clamped shifts are interpolated to calculation frequencies and bounded by the physical attenuation limit of the specified coupling (V_{floor}) and receiver saturation limits (80 dB):

$$G(f) = \max(V_{floor}, \min(80, G_{heuristic}(f) + \delta_{clamped}(f))) \quad (45)$$

4. Downstream Objective Evaluation: The resulting clamped gain vector $\mathbf{G}$ is then routed through the C++ engine to compute the final desensitized SII and evaluate all remaining acoustic and physiological penalties (e.g., P_{loud} , P_{cr}).

Loss Function Formulation

The Nelder-Mead solver minimizes:

$$\mathcal{L}(\boldsymbol{\delta}) = -100 \cdot \text{SII}_{desens}(\mathbf{G}) + P_{anchor} + P_{bounds} + P_{loud} + P_{spl} + P_{rough} + P_{order} + P_{cr} + P_{abg} \quad (46)$$

where SII_{desens} is the effective Speech Intelligibility Index calculated using the

699 johnson2011_smoothed transfer function.

700 **Explicit Penalty Terms and Exact Weights (λ)**

701 **1. Anchor Penalty ($\lambda_{anchor} = 0.1$):**

$$P_{anchor} = 0.1 \sum_{j=1}^6 |\delta_j| \quad (47)$$

702 **2. Out-of-Bounds Penalty ($\lambda_{bounds} = 1000.0$):**

$$P_{bounds} = 1000.0 \left[\sum_{j=1}^6 \max(0, \delta_j - 30)^2 + \sum_{j=1}^6 \max(0, -\delta_j - 60)^2 \right] \quad (48)$$

703 **3. Physiological Loudness Ceiling Penalty ($\lambda_{loud} = 2000.0$):**

$$P_{loud} = 2000.0 \cdot \max(0, \text{Sones}_{MG04}(\mathbf{G}) - \text{Cap}) \quad (49)$$

704 *(Note: While gain-based penalties in this framework are squared to strongly penalize large*
705 *deviations, the loudness penalty is explicitly linear. This is a deliberate choice: because*
706 *sones inherently represent a compressive power-law transformation of physical acoustic*
707 *energy, a linear penalty in the sone domain naturally exerts an exponentially growing*
708 *restriction on the underlying insertion gain. Squaring the sone error introduces severe*
709 *mathematical stiffness and destabilizes the simplex gradient. Furthermore, coupling a*
710 *linear penalty slope of $\lambda_{loud} = 2000.0$ against the intelligibility objective term ($-100 \cdot SII$)*
711 *means that violating the cap requires a marginal objective benefit of $\partial SII / \partial S > 20 \text{ sone}^{-1}$.*

Because the SII is strictly bounded between 0 and 1, this condition is physically impossible to satisfy, turning the soft penalty into a rigorous, guaranteed mathematical boundary.)

where Sones_{MG04} is the Moore & Glasberg (2004) specific-loudness integration computed via native C++. The dynamic U-shaped cap is interpolated across Pure Tone Average knots $\mathbf{PTA}_{knots} = [10, 32.5, 52.5, 72.5, 90]$ dB HL from level-specific monaural sone vectors (note: these vectors represent single-ear monaural limits, which equal exactly half of their binaural equivalents under the simple-doubling convention of the 2004 framework):

- $L_{in} = 50$ dB SPL: $\mathbf{K}_{50} = [1.5, 1.0, 0.8, 1.2, 1.2]$ monaural sones
- $L_{in} = 65$ dB SPL: $\mathbf{K}_{65} = [7.0, 5.0, 5.5, 6.5, 6.5]$ monaural sones
- $L_{in} = 80$ dB SPL: $\mathbf{K}_{80} = [20.0, 12.0, 10.0, 15.0, 14.0]$ monaural sones

$$\text{Cap}_{base} = \text{interp}(\text{PTA}_{sn}, \mathbf{PTA}_{knots}, \mathbf{K}_{L_{in}}) \quad (50)$$

where PTA_{sn} is defined as the four-frequency pure-tone average of the sensorineural component at 500, 1000, 2000, and 4000 Hz.

Profile Adjustments:

- Reverse-slope restriction ($L_{in} \geq 75$ dB SPL and low-to-high threshold difference > 10 dB): $\text{Cap} = \text{Cap}_{base} - 0.10 \cdot (\overline{\text{HTL}}_{\leq 500} - \overline{\text{HTL}}_{\geq 4000})$.
- Conductive air-bone gap adjustment: $\text{Cap} \leftarrow \text{Cap} - 0.25 \cdot \text{PTA}_{ABG}$ (if $L_{in} \geq 75$ dB SPL) or $+0.10 \cdot \text{PTA}_{ABG}$ (if $L_{in} < 75$ dB SPL).

4. **Broadband SPL Saturation Ceiling Penalty** ($\lambda_{spl} = 2000.0$):

$$P_{spl} = 2000.0 \cdot \max(0, \text{SPL}_{aided} - 110.0) \quad (51)$$

(Note: This 110 dB SPL penalty evaluates the summed broadband RMS power of the entire amplified signal to enforce an overall physiological safety limit. It operates independently of the 120 dB SPL Maximum Power Output (MPO) ceiling defined in Eq. 43, which dictates the absolute hardware saturation threshold for individual narrow bands. For example, while multiple individual frequency bands may operate safely below their respective 120 dB SPL MPO limits, their combined acoustic energy can still sum to a broadband level that triggers this 110 dB SPL overall penalty, ensuring aggregate exposure remains bounded.)

5. **Spectral Roughness Penalty** ($\lambda_{rough} = 0.5$):

$$P_{rough} = 0.5 \sum_{j=1}^5 (\delta_{j+1} - \delta_j)^2 \quad (52)$$

6. **Inter-Level Order Monotonicity Penalty** ($\lambda_{order} = 2000.0$): Enforces $G_{50} \geq G_{65}$

and $G_{80} \leq G_{65}$:

$$P_{order} = \begin{cases} 2000.0 \sum_{j=1}^6 \max(0, G_{65,j} - G_{50,j})^2, & \text{for } L_{in} = 50 \\ 2000.0 \sum_{j=1}^6 \max(0, G_{80,j} - G_{65,j})^2, & \text{for } L_{in} = 80 \end{cases} \quad (53)$$

7. Compression Ratio Ceiling Penalty ($\lambda_{cr} = 200.0$):

$$P_{cr} = \begin{cases} 200.0 \sum_{j=1}^6 \max(0, (G_{50,j} - G_{65,j}) - \Delta_{max,j})^2, & \text{for } L_{in} = 50 \\ 200.0 \sum_{j=1}^6 \max(0, (G_{65,j} - G_{80,j}) - \Delta_{max,j})^2, & \text{for } L_{in} = 80 \end{cases} \quad (54)$$

where $\Delta_{max,j} = 10.0 \cdot (1 - \text{ABG}_j / \max(0.001, \text{HTL}_j))$ for soft speech, bounding emergent compression ratios safely below 3.0:1 for sensorineural loss while enforcing linear amplification for conductive components. This 3.0:1 ceiling represents the best-supported parameter in the algorithm, firmly grounded in empirical psychoacoustic literature (Souza, 2002; Souza et al., 2006) demonstrating severe envelope flattening, loss of acoustic contrast, and speech-in-noise deficits for CRs $> 3.0 : 1$. Because this is enforced as a soft penalty ($\lambda_{cr} = 200.0$) rather than a hard algorithmic clamp, the solver can mathematically violate it when pushed against even stronger boundaries (e.g., yielding CRs > 3.0 in profound losses like A4 or A5), effectively transitioning from wide dynamic range compression to hard limiting.

8. Air-Bone Gap Excursion Penalty ($\lambda_{abg} = 1.0$):

$$P_{abg} = \begin{cases} 1.0 \sum_{j=1}^6 \max(0, \delta_j)^2, & \text{if } \exists j, \text{Loss}_j > 0 \\ 0, & \text{otherwise} \end{cases} \quad (55)$$

Solver Controls and Convergence Tolerances

- **Algorithm:** Nelder-Mead Simplex via R's `stats::optim()`.
- **Iteration Ceiling:** `control = list(maxit = 800)`.

- **Convergence Tolerance:** Relative convergence tolerance `reltol = sqrt(.Machine$double.eps)`
 $\approx 1.49 \times 10^{-8}$.
- **Initial Simplex Seeding:** Seeding incorporates an audibility projection
 $\delta_{start} = \min(20, \max(0, \text{target_aided} - G_{heuristic}))$, where $\text{target_aided} =$
 $\min(\text{UCL} - 5, \max(L_{in}, \text{HTL} + 10))$. Soft inputs receive +3 dB shift; loud
inputs receive $\max(-10, \delta_{start} - 5)$ dB shift.
- **Multi-Start Strategy:** A 3-iteration multi-start routine (seeding the initial simplex
with the NAL-R target and executing two additional randomized restarts with uniform
random jitter $\delta_{start} \leftarrow \delta_{start} + \mathcal{U}(-5, +5)$) is deployed to stabilize numerical convergence,
though Nelder-Mead inherently lacks formal global convergence guarantees.

S.II. Full Objective Reference Tables

The following tables provide the complete prescriptive target data for all seven reference profiles (A1-A7; Johnson & Dillon, 2011) evaluated at 65 dB SPL. These tables report canonical single-ear monaural loudness calculations exclusively (binaural equivalence requires propagating the dynamic full-range signal through a non-linear binaural loudness engine).

TABLE S5. Monaural Loudness (Sones) and Desensitized SII across A1-A7 Audiograms (65 dB SPL Input).

Profile	Formula	Monaural Loudness (sones)	Desensitized SII
A1	NAL-NL2	3.99	0.73
	Open-NL (Occluded)	4.44	0.81

Profile	Formula	Monaural Loudness (sones)	Desensitized SII
A2	Open-NL (Vented)	4.43	0.78
	NAL-NL2	3.10	0.77
	Open-NL (Occluded)	4.44	0.83
A3	Open-NL (Vented)	4.45	0.84
	NAL-NL2	3.51	0.61
	Open-NL (Occluded)	4.28	0.76
A4	Open-NL (Vented)	4.28	0.72
	NAL-NL2	5.96	0.67
	Open-NL (Occluded)	4.78	0.76
A5	Open-NL (Vented)	5.21	0.56
	NAL-NL2	5.36	0.54
	Open-NL (Occluded)	4.24	0.65
A6	Open-NL (Vented)	4.26	0.46
	NAL-NL2	2.32	0.78
	Open-NL (Occluded)	3.33	0.82
A7	Open-NL (Vented)	3.32	0.82
	NAL-NL2	1.15	0.98
	Open-NL (Occluded)	1.63	0.96
	Open-NL (Vented)	1.63	0.96

TABLE S6. Insertion Gain Targets (dB) across A1-A7 Audiograms (65 dB

775

Profile	Formula	250 Hz	500 Hz	1000 Hz	2000 Hz	4000 Hz	8000 Hz
A1	NAL- NL2	0.0	0.0	7.3	12.1	18.0	19.1
A1	Open-NL (Oc- cluded)	-0.2	-5.5	-3.6	7.6	22.2	14.0
	Open-NL (Vented)	-2.6	-4.8	-2.7	8.0	22.0	12.7
A2	NAL- NL2	10.1	9.3	12.2	8.3	3.9	4.0
A2	Open-NL (Oc- cluded)	-2.3	-5.9	6.4	17.5	26.2	23.4
	Open-NL (Vented)	-0.5	-5.0	5.1	17.6	26.2	13.8
A3	NAL- NL2	0.0	0.0	9.9	16.8	20.7	21.6
A3	Open-NL (Oc- cluded)	-3.7	-5.7	17.4	36.0	36.8	33.8

Profile	Formula	250 Hz	500 Hz	1000 Hz	2000 Hz	4000 Hz	8000 Hz
	Open-NL (Vented)	-5.4	-6.1	14.9	37.3	40.8	19.5
A4	NAL- NL2	0.0	0.0	10.5	18.0	21.8	21.8
A4	Open-NL (Oc- cluded)	-5.3	-10.0	10.2	36.0	48.8	23.0
	Open-NL (Vented)	-5.2	-8.7	7.3	39.4	41.4	29.8
A5	NAL- NL2	0.0	0.0	14.9	24.3	27.5	27.5
A5	Open-NL (Oc- cluded)	12.7	21.4	34.9	48.9	41.4	23.0
	Open-NL (Vented)	18.5	18.8	34.1	49.1	41.7	23.8
A6	NAL- NL2	18.5	18.5	24.0	25.7	25.7	11.6
A6	Open-NL (Oc- cluded)	34.6	38.9	43.9	47.0	50.5	49.6

Profile	Formula	250 Hz	500 Hz	1000 Hz	2000 Hz	4000 Hz	8000 Hz
	Open-NL (Vented)	0.0	11.0	29.0	45.1	50.5	49.6
A7	NAL- NL2	34.7	34.7	34.7	36.6	38.0	35.1
A7	Open-NL (Oc- cluded)	34.6	38.9	40.2	39.5	39.2	34.6
	Open-NL (Vented)	0.0	11.0	25.4	37.6	39.2	34.6

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